

Measuring the Simulation Gap in Root Cause Analysis for CNC Milling: A Physics-Based Simulator Grounded in Real Multi-Sensor Data

Author names and affiliations to be added

DRAFT — target venue not yet chosen; several results are incomplete (see Section 6)

Abstract

Root cause analysis (RCA) of machining faults lacks public data with ground-truth causes, so RCA methods are routinely evaluated on simulators whose generator is also used to build the diagnostic nodes. We build a regenerative-milling simulator (delay-differential equations with helical flutes and edge forces) and ground its healthy baseline in a real multi-sensor milling dataset by modelling accelerometer and microphone spectra and their dependence on cutting conditions. Fault effects (stiffness, damping, tool wear, depth, spindle speed) are injected through the simulator and scaled by an effect-size calibration against real anomalous cuts. We then evaluate six causal RCA algorithms on a 14-node graph following a root-cause-ranking protocol. When diagnostic nodes and test faults come from the same generator, top-1 accuracy at $m=10$ interventional samples is 0.52–0.98 (0.80–0.98 for five of six algorithms); when the simulator’s physical parameters are shifted at test time it falls to 0.35–0.71, and the ordering of methods changes. The simulated healthy baseline is still distinguishable from real data (classifier two-sample test, balanced accuracy 0.82 ± 0.04), and no real root-cause validation was completed. We therefore present the results as a measurement of how strongly simulated RCA benchmarks depend on generator assumptions, not as evidence of real-world diagnostic accuracy.

1 Introduction

Diagnosing why a milling operation misbehaves (loose fixture, worn tool, excessive depth, unsuitable spindle speed) requires knowing which physical cause produced the sensor signals. Real datasets rarely record causes; they record sensor streams and, at best, expert labels of “normal” or “abnormal”. The recently released MSM dataset [5] is a rare multi-sensor milling collection with expert labels, but it does not provide root-cause annotations for most anomalies. Causal RCA methods, several developed for microservices [3, 7] and related outlier-explanation settings [6], take an observational dataset of healthy behaviour and a few anomalous samples, and rank graph nodes by how much their mechanism changed. Applying them to machining needs a source of faulty data with known causes, and the natural source is a simulator.

The problem. A simulator-based RCA benchmark is only informative if the simulated data resemble real data and if the diagnostic procedure does not exploit the simulator itself. Both conditions are hard to check: real root causes are unavailable, and nodes of the causal graph are usually derived from, or trained on, the same generator that produces the test faults.

What we do. (1) We implement a regenerative-milling simulator for the tool used in MSM, with every parameter tagged as fixed from a catalogue, learned from data, or assumed (Appendix A). (2) We build the healthy baseline from real spectra rather than from the simulator, and report honestly how close it is (classifier two-sample test). (3) We inject simulated faults and calibrate their scale by effect size on real anomalous cuts. (4) We run six causal RCA algorithms and test them with and without a mismatch between the generator used to build the nodes and the one used to generate the faults.

Main finding. The accuracy that causal RCA methods reach in this benchmark depends strongly on whether the generator is shared: top-1 accuracy at $m=10$ drops from 0.80–0.98 (five of six algorithms) to 0.35–0.71 (all six) under moderate shifts of the simulator’s stiffness, damping and force coefficients, and the ranking of the algorithms changes.

What we do not show. We have no validation on real root causes, the simulated healthy baseline is distinguishable from real data, and the ablations that would separate the effect of the causal graph from that of the diagnostic nodes were not completed (Section 6).

2 Related Work

Regenerative chatter and milling simulation. Regenerative chatter stability and its analytical and time-domain treatment are surveyed by Altintas et al. [1]; we use the zero-order stability-lobe approximation and a time-domain delay model in that tradition.

Machining datasets. Kim et al. [5] release synchronised accelerometer, microphone, current and controller data from three machines with expert anomaly labels (normal, abnormal, defective tool). We use a subset (Section 3).

Causal root cause analysis. RCD [3] localises root causes with conditional-independence tests, BARO [7] uses multivariate Bayesian online change-point detection, and Orchard et al. [6] study root causes of outliers without full structural knowledge; our “Smooth Traversal” baseline follows the repository’s description attributed to this work [VERIFY: confirm that the implemented algorithm matches the paper]. BRCD, RCG and SimpleRCA are implemented in our codebase from the repository’s descriptions [CITATION NEEDED: BRCD (Lee, Zhou, Kocaoglu, ICML 2026), RCG (UAI 2025), SimpleRCA (Fang 2025): the author confirms the sources; bibliographic entries still to be added].

Calibrating simulators to data. Calibration of computer models with a model-discrepancy term is classical [4]; domain randomisation [9], adversarial refinement of simulated data [8] and simulation-based inference [2] are alternatives. We considered adversarial refinement but did not use it because of the small number of real cuts; we use a classifier two-sample test only as a diagnostic [CITATION NEEDED: Lopez-Paz and Oquab 2017; Gretton et al. 2012].

3 Data

We use datasets 5 and 6 of machine `imi_vm20i` (Hurco, wet cutting, AA 6061): tool 2SE0250IX150A (2 flutes, 6.35 mm, flute length 38.1 mm, per the manufacturer catalogue), axial depth 38.1 mm,

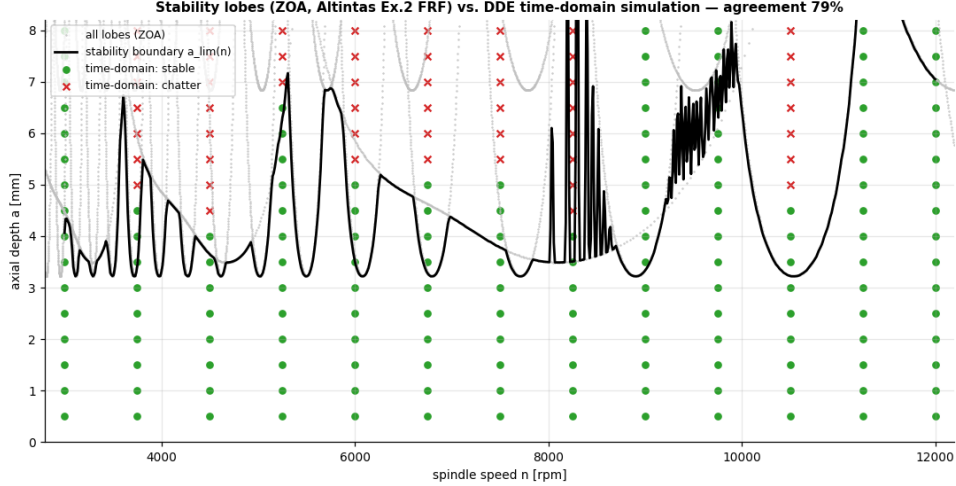


Figure 1: Stability lobes (zero-order approximation, 4-flute reference transfer function) and time-domain simulation outcomes on a speed–depth grid.

spindle speed 9000–12000 rpm, radial depth $a_e \in \{0.0635, 0.1905, 0.3175\}$ mm, up and down milling. This gives 73 cuts (48 labelled normal, 25 abnormal); each cut is split into three 0.3 s windows (219 windows). We verified that the spindle speed and feed in the label files equal the controller-recorded values (ratio 1.000). In these two datasets the abnormal label is strongly associated with up-milling rather than with a specific root cause (all 36 down-milling cuts are normal). Cuts with a documented root cause (defective tools, clamp failure, excessive depth) exist in other datasets but use different tools, and we could not complete their analysis (Section 6).

4 Method

4.1 Simulator

The tool-tip displacement follows modal dynamics driven by cutting forces; the chip thickness of each flute slice depends on the current and delayed displacement. For each of the $N_z=8$ axial slices of a flute, with helix lag $\psi(z) = 2z \tan \beta / D$, the elemental forces are

$$dF_t = (K_t h + K_{te}) dz, \quad dF_r = (K_r K_t h + K_{re}) dz, \quad (1)$$

where h is the chip thickness (clipped at 0 and a cap), K_{te}, K_{re} are edge-force coefficients and $\beta=30^\circ$ (manufacturer series feature list; medium confidence). Spindle imbalance adds a synchronous rotating force $U\Omega^2$. With helix and edge forces switched off the implementation reproduces our earlier straight-flute reference to 7×10^{-16} relative error. The zero-order stability-lobe computation reproduces the repository’s previous implementation exactly on the same transfer function, and a time-domain grid agrees with the lobe prediction on 79% of (n, a) points (Figure 1); the disagreement is concentrated at moderate depth, where the zero-order approximation is known to be conservative for half-immersion cuts.

4.2 Provenance of every parameter

Following a fixed policy, a parameter is *fixed* if a verified source gives its value for this exact tool and context, taken from a *range* if a source gives one, and *learned* from data otherwise; values with no

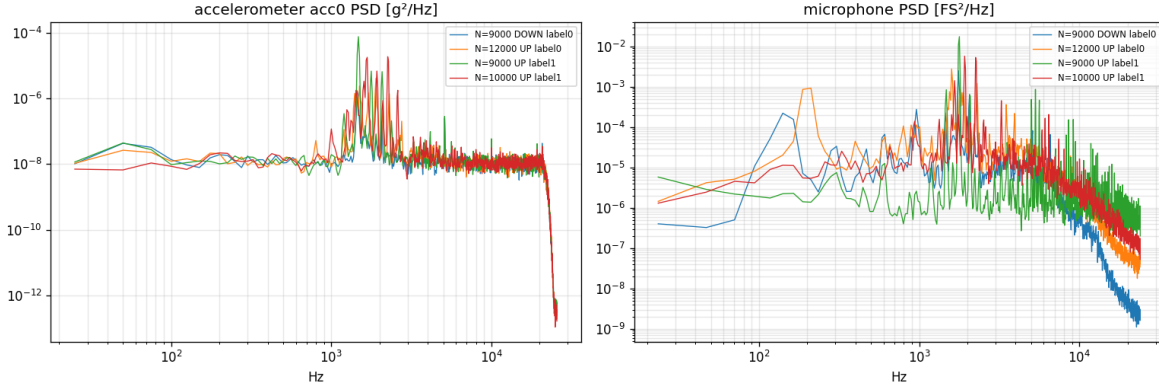


Figure 2: Power spectral density of four real windows (accelerometer left, microphone right): normal and abnormal cuts at different spindle speeds.

source or data are marked *assumed*. Appendix A lists each parameter, its tier and the verification status. Cutting-force coefficients for this tool and material, the tool-tip transfer function and the spindle imbalance had no usable source in our search and are therefore not fixed; we found no verified range for them either. In practice our direct attempts to learn them failed (Section 5.2), so the simulator’s physical parameters are nominal values, which is itself an assumption.

4.3 Healthy baseline from real spectra

For each 0.3 s window we estimate the one-sided power spectral density of the two accelerometer channels and the microphone, and summarise it by a piecewise-linear envelope over log-frequency (18 knots; knot values are band-averaged linear power so that total energy is preserved) plus the rms amplitude of the first eight harmonics of the spindle frequency $f_0 = n/60$. The parameters are regressed on the cutting conditions (kernel ridge on $\log n$, $\log a_e$, $\log f$, direction, label) and the residuals are decomposed by one-way variance analysis into a between-cut and a within-cut component on twelve principal directions, with out-of-sample residuals. Synthetic windows are generated by shaping complex Gaussian spectra with the envelope and adding harmonics with random phases. The real spectra differ clearly from white or pink noise: the accelerometer shows a flat floor, a brick-wall cut-off near 23 kHz and a cluster of peaks at 1.5–2.3 kHz that is about two orders of magnitude stronger in abnormal windows; the microphone shows a broad hump at 1–6 kHz (Figure 2).

4.4 Fault injection and gain calibration

For each window and fault class we run the simulator twice at identical conditions and noise seed (healthy and faulty), and add the scaled difference $g(s_{\text{fault}} - s_{\text{healthy}})$ to the real-spectra baseline, with separate gains g_{acc} , g_{mic} . The five faults and their severity ranges (assumed, no source) are: stiffness $\times [0.4, 0.9]$; damping $\times [0.3, 0.8]$; tool wear $K_t \times [1.1, 1.6]$ with edge forces increased; radial depth $\times [1.5, 4]$; spindle speed $\pm [3\%, 15\%]$. Matching the distribution of simulated faulty features to real abnormal windows did not identify the gains (the objective was flat for small gains, Section 5.2); we therefore set them by effect size, so that the mean shift of $\log_{10}$ acceleration rms and of microphone level equals the real difference between abnormal and normal windows (+0.375 and +4.86 dB), giving $g_{\text{acc}} = 7.7 \times 10^{-6}$ and $g_{\text{mic}} = 0.0116$. Because real abnormal windows here are mainly up-milling cuts, this scale is an assumption about how strong a fault is, not an estimate of it.

Real-noise model variant	C2ST balanced accuracy (0.5 = indistinguishable)
Linear regression, mean only	0.986
Linear regression + between/within-cut variation	0.876
Kernel ridge + stratified split, mean only (3 splits)	0.864 ± 0.022
Kernel ridge + hierarchical variation (3 splits)	0.820 ± 0.036

Table 1: Classifier two-sample test between real and simulated healthy windows on held-out cuts. Three-split values are mean $\pm$ standard deviation over random stratified splits. [\[VERIFY: values copied from console logs; only the last run is archived\]](#)

4.5 Causal RCA protocol

We follow the root-cause-ranking protocol of the repository: a healthy dataset D_{obs} and m anomalous samples D_{int} are given, and each algorithm ranks all 14 nodes; we report top- k accuracy for the true root-cause node. The nodes are five root-cause nodes (estimated stiffness, damping, tool wear, depth, and a controller-measured spindle-speed deviation) and nine sensor nodes (log acceleration, microphone level, two spectral non-synchronous indices, three accelerometer bands, two microphone bands). The directed graph encodes physical prior knowledge (each root cause points to the sensor quantities it can affect). The first four root-cause nodes are soft sensors: gradient-boosted regressors of the logarithm of the latent fault factor from the 30 spectral features, trained on training cuts only (out-of-fold for D_{obs}); the spindle-speed node is the true factor plus 0.5% reading noise. All nodes are standardised with D_{obs} . Because the soft sensors are trained on labels from the same simulator, we add a *shifted* test where the simulator’s natural frequency, damping, stiffness, force and edge coefficients are multiplied by random factors in $[0.75, 1.33]$, $[0.5, 2]$, $[0.6, 1.7]$, $[0.6, 1.6]$ and $[0.4, 2.5]$ respectively for each cut, while the soft sensors keep their nominal-parameter training. Cuts are split by identity into 43 training and 30 test cuts, stratified by a_e ; each test point averages 100 random draws of m faulty windows per fault.

5 Experiments

5.1 How realistic is the healthy baseline?

We train the baseline model on a subset of cuts and test it on held-out cuts, using a random-forest classifier two-sample test on the four summary features and the conditions, with balanced accuracy and fold assignment by cut (0.5 means indistinguishable). The first version of this metric used plain accuracy with five synthetic windows per real window, whose chance level is 0.83; those numbers are discarded. Table 1 shows the progression; the final variants remain far from 0.5 and the differences between the kernel-regression and hierarchical variants are within the spread across data splits. Feature means match well (e.g. log acceleration -1.37 simulated vs. -1.38 real), but their dispersion and joint dependence on conditions do not. We could not measure an empirical floor for this test because no cutting condition is repeated in the data (only three a_e values and one cut per condition).

5.2 A negative result: direct calibration of the delay model

We tried to fit the simulator’s physical and sensor parameters directly to real windows with CMA-ES, matching feature distributions (MMD), later matching group means over cutting conditions. With 15 free parameters the loss plateaued near 3 (squared errors in units of the real standard

Classifier	Top-1 $\uparrow$	Top-3 $\uparrow$	Macro-F1 $\uparrow$	AUROC (fault vs. normal) $\uparrow$
Gradient boosting	0.886	0.990	0.885	1.000
Random forest	0.870	0.990	0.868	0.998
MLP	0.807	0.987	0.804	0.994
QDA (Bayes)	0.787	0.978	0.779	0.992
Logistic regression	0.726	0.966	0.722	0.990
kNN	0.722	0.922	0.717	0.983
Gaussian Naive Bayes	0.579	0.912	0.554	0.961

Table 2: Six-class classification on simulated data, 16 held-out cuts (chance top-1 = 0.167).

deviation) and did not converge after 35–40 generations; we stopped it by a pre-set rule. Earlier low-MMD fits coexisted with qualitatively wrong dependence on conditions: in the real data the microphone level correlates with spindle speed at +0.85 while the simulation gave 0.00, and the simulated acceleration depended on radial depth (0.69) where the real one did not (0.06). A bug that we had to fix on the way is instructive: real features had been computed on windows of about 4 s and simulated features on 0.18 s, which inflated simulated variance threefold. We conclude that the available data do not constrain the delay model’s absolute levels, which is why we use the simulator only for fault *differences*.

5.3 Classifying simulated fault classes

Table 2 reports six-class classification (healthy plus five faults) on 16 held-out cuts, with hyperparameters chosen on 14 validation cuts. Tree ensembles are best; Gaussian generative models are worse, consistent with highly correlated continuous features. These numbers hold *within the simulator*. Applying the models to real windows, 91–97% of real normal windows are classified normal by the three tree and neighbour models, but only 51% and 57% by logistic regression and Naive Bayes, and 61, 64 and 72 of 75 real abnormal windows are assigned to tool wear by QDA, random forest and gradient boosting respectively. There is no ground truth for the abnormal windows, so we read this as a symptom of the models mapping any broadband increase to tool wear, not as a diagnosis.

5.4 Causal RCA

Table 3 gives mean top-1 accuracy over the five root causes (chance = $1/14 \approx 0.07$). With a shared generator, Smooth Traversal and BARO approach 1.0 as m grows, BRCD and RCG reach about 0.85–0.90, RCD declines with m , and SimpleRCA stays near 0.5. Under shifted simulator parameters all methods drop; the ordering changes: BARO falls from 0.94 to 0.35 at $m=10$ and declines with m , whereas BRCD and RCG improve slowly with m and Smooth Traversal remains best at 0.71. In the shifted setting top-3 accuracy at $m=10$ is 0.98 for Smooth Traversal, 0.93 for BARO and 0.89 for BRCD (Figure 3). Several entries equal 1.00 in the shared-generator setting; we do not regard them as evidence of diagnostic ability because the spindle-speed node contains the answer by construction and the other root-cause nodes are supervised estimates of it (Section 6).

5.5 Dependence on the assumed fault strength

Scaling the fault gains by 0.3 and 3 changes the best classifier’s accuracy from 0.698 to 0.943 (Table 4). Since the gain is set by an assumption, any single accuracy figure from this benchmark should be read together with this curve.

Algorithm	Same generator (in-distribution)				Shifted DDE parameters			
	$m=5$	10	20	50	$m=5$	10	20	50
BRCD	0.87	0.90	0.89	0.85	0.59	0.58	0.62	0.64
RCD	0.79	0.80	0.71	0.59	0.45	0.39	0.30	0.21
RCG	0.87	0.90	0.89	0.85	0.58	0.59	0.63	0.64
SmoothTraversal	0.95	0.98	1.00	1.00	0.68	0.71	0.77	0.81
BARO	0.88	0.94	0.98	1.00	0.42	0.35	0.30	0.26
SimpleRCA	0.53	0.52	0.52	0.48	0.42	0.44	0.44	0.40

Table 3: Causal RCA: top-1 accuracy (mean over 5 faults; 100 draws per fault) versus number of anomalous samples m .

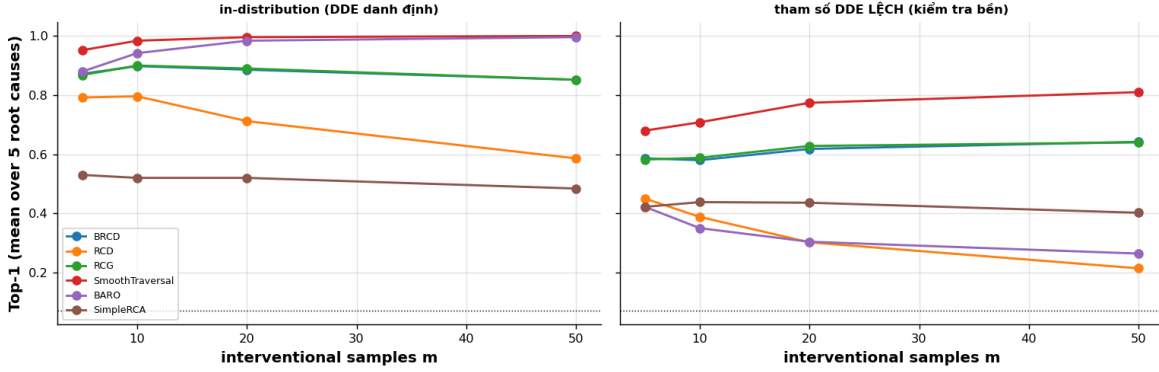


Figure 3: Top-1 accuracy of the causal algorithms versus m for the shared-generator and shifted settings (dotted line: chance).

6 Limitations and What Remains

- **No real root-cause validation.** Real cuts with documented causes (defective tools, clamp failures, depth) are on other tools; the comparison of a defective tool with a healthy one of the same type was planned but the data download was interrupted. **[TODO: run real_defect_check on datasets 3 and 14.]**
- **Circularity.** The soft sensors are trained on simulator labels, the spindle-speed node uses the latent factor, and the causal graph was written to match the fault model. The shifted setting reduces but does not remove this. A label-free node variant, shuffled and empty graphs, and three-seed confidence intervals were implemented but not run to completion. **[TODO: run rca_ablation.]**
- **Baseline realism.** The healthy baseline is distinguishable from real data (balanced accuracy 0.78–0.90 across variants and splits) and we could not measure a floor.
- **Assumed quantities.** Fault severity ranges, gains, nominal physical parameters and the causal graph are assumptions; force coefficients and tool-tip dynamics for this tool had no verified source.
- **Scope.** One tool, one workpiece material, one machine, 73 cuts, mostly up- versus down-milling as the real abnormal signal.

Classifier	gain $\times 0.3$	$\times 1$ (calibrated)	$\times 3$
Gradient boosting	0.698	0.882	0.943
Random forest	0.674	0.855	0.922
QDA (Bayes)	0.545	0.769	0.847
Logistic regression	0.535	0.729	0.834

Table 4: Top-1 accuracy of classifiers on simulated data versus fault gain relative to the calibrated value.

- **Single split.** The RCA results use one cut split and one generation seed and carry no confidence intervals.
- **Open hypothesis.** We suspect that the 1.5–2.3 kHz cluster is a tool-structure mode excited when a harmonic of the tooth-passing frequency is close to it; adding a harmonic-proximity feature did not give a clear improvement.

7 Conclusion

A simulator grounded in real healthy spectra lets us run causal RCA algorithms on machining faults with known causes, but the resulting accuracy depends on how much the diagnostic nodes share with the generator, and falls from near 1.0 to 0.35–0.71 under moderate mismatch of the simulator’s physical parameters. This is a measurement of the dependence of a simulated benchmark on its assumptions, not a claim about real machining diagnosis, which requires the real-cause validation and ablations listed above.

Code and data. Code: `cutting_sim/` in the project repository (branch `cutting-sim-experiments`); data: MSM [5] (CC BY 4.0). **[TODO: release instructions]**

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A Parameter provenance

B Methodological pitfalls found along the way

1. Plain accuracy of a classifier two-sample test with unequal sample sizes has a chance level above 0.5; use balanced accuracy.
2. Real and simulated features must be computed on windows of identical length; a 20-fold mismatch changed the apparent variance threefold.
3. Out-of-sample residuals are needed to estimate between-cut variation when the regression can interpolate each cut.
4. Distribution matching alone did not identify the fault gain; effect-size matching was used instead.
5. Nodes of a causal graph that are trained on the generator’s labels make the benchmark circular.

Parameter	Tier	Verification
tool diameter, flutes, flute length, overall length	FIXED	verified (read from catalog table p.5)
helix angle β	FIXED	medium: from series feature list (p.3); not stated per part number
tool substrate/coating	FIXED	medium
N, ae, feed F, direction	FIXED	verified (N and F equal mtc Spindle_Speed/Feed_Rate, ratio 1.000)
axial depth ap = engaged depth	FIXED	assumption (not measured)
workpiece AA 6061; wet cutting (coolant flood)	FIXED	verified (stated in paper)
sampling: accelerometer 51.2 kHz, audio 48 kHz	FIXED	verified (stated in paper)
analysis window 0.3 s	FIXED	design choice
model structure: regenerative DDE, shear+edge force, ZOA lobes	FIXED	model structure
Kt, Kr, Kte, Kre (AA 6061, this tool, wet)	LEARNED	NO usable source (rejected: steel-like values; dry-cutting AA6061 value unverified)
tool-tip FRF: fn_x, fn_y, ζ , modal stiffness k	LEARNED	NO source for this tool/holder/stick-out (Altintas Ex.2 FRF rejected: different machine and tool)
spindle imbalance U	LEARNED	NO source (rotor mass, balance grade unknown)
sensor gains, noise floors, rpm-dependence	LEARNED	NO source (sensors uncalibrated: mic dBFS, accelerometer mm/s ²)
runout; between-cut variability σ (Kt, runout, roughness)	LEARNED	NO source
chip-thickness cap 0.3 mm; roughness noise base 0.02 μ m	ASSUMED	own assumption

Table 5: Every simulator parameter, its tier (fixed / learned / assumed) and verification status.